Structural Breaks in Blockchain/Diffusion Ledger Technology Research within Banking Literature: A Bibliometric Study with Endogenous Break Detection

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Abstract

This study examines the publication dynamics of blockchain and distributed ledger technology (DLT) research within the banking literature using a large-scale bibliometric framework. Drawing on 498,958 banking-domain articles from Semantic Scholar (2000–2025), we identify 7,280 papers classified by 12 word-boundary stems and apply endogenous structural break detection to the yearly publication series. The supremum F-test (sup$F$) reveals a single significant structural break in 2012 ($F = 21.5$, wild-bootstrap $p = 0.002$), coinciding with Bitcoin's first mainstream attention cycle. No later breaks are detected at 2017, 2020, or 2022, indicating that post-2017 growth represents an exponential continuation of the pre-2012 trend rather than a new regime shift. Venue tier analysis shows 93–96% of blockchain papers appear in established venues across all eras, with preprint and rapid-review outlets never exceeding 5.3%. A pooled classification audit yields 60% precision (Wilson CI: 50–69%), declining from 74% pre-2017 to 40% post-2022 due to inflation from generic fintech/AI papers mentioning blockchain peripherally. Cross-wave correlation analysis reveals significant co-movement between AI and blockchain waves ($r = 0.476$, $p = 0.014$), while blockchain–green finance co-movement is negligible ($r = 0.104$, $p = 0.61$). These findings demonstrate that blockchain diffusion in banking follows a single early-inflection growth curve with stable venue composition, distinguishing it from hype-cycle narratives and providing methodological advances for structural break analysis in bibliometric studies.

Keywords: blockchain, distributed ledger technology, bibliometrics,

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structural break detection, supF test, banking literature, publication dynamics, technology diffusion

1. Introduction

Blockchain and distributed ledger technologies (DLT) have attracted unprecedented research attention since the publication of Nakamoto's whitepaper on Bitcoin in 2008 [1]. The banking industry, as both an early adopter and a primary target of blockchain disruption, has experienced a particularly rapid expansion of related scholarship. From fewer than 10 papers per year before 2013, blockchain-related publications within the banking literature grew to over 2,100 papers by 2025, constituting approximately 6.2% of all banking-domain research output.

Despite this explosive growth, several fundamental questions remain unanswered. First, has blockchain research in banking experienced genuine structural breaks in its publication trajectory—that is, discrete shifts in the underlying growth regime—or has growth been continuous? Prior bibliometric studies of blockchain have documented output trends [2, 3] but have not applied formal structural break tests to the publication series. Second, does the post-2017 cryptocurrency boom represent a new publication regime, or does it simply extend the pre-existing growth curve? Third, has the rapid expansion of blockchain research been accompanied by a shift toward lower-quality venues—a pattern hypothesized but not systematically tested.

This study addresses these gaps using a multi-method bibliometric framework applied to a corpus of 498,958 banking-domain articles. We employ the supremum F-test (sup$F$) of **(author?)** [4] with wild-bootstrap p-values to detect endogenous structural breaks in the blockchain publication series. We complement this with Mann-Kendall trend testing [5], Sen's slope estimation [6], and era-stratified venue tier classification. Additionally, we analyze co-movement between the blockchain wave and parallel technology waves (artificial intelligence, green finance) using detrended cross-correlation analysis.

Our contributions are threefold. Methodologically, we demonstrate the application of endogenous break detection—developed for macroeconomic time series—to bibliometric publication data, revealing a single break year (2012) rather than the multiple regime shifts hypothesized by hype-cycle narratives. Substantively, we show that the 2012 break coincides with Bitcoin's first mainstream attention cycle and that subsequent growth is a continuous

exponential trajectory rather than a new regime. Finally, we provide the first systematic evidence that venue tier composition in blockchain research remains stable across eras, challenging assumptions that publication growth is accompanied by venue degradation.

The remainder of this paper is organized as follows. Section 2 reviews related work across four domains. Section 3 describes the data, classification system, audit methodology, and statistical tests. Section 4 presents the findings organized by growth magnitudes, break detection, venue composition, audit quality, and cross-wave co-movement. Section 5 interprets these findings in relation to existing literature and discusses limitations. Section 6 concludes with implications for bibliometric methodology.

2. Related Work

2.1. Blockchain Bibliometrics

The growing body of blockchain research has itself become the subject of bibliometric investigation. **(author?)** [2] conducted a comprehensive systematic review of blockchain research in the banking sector, identifying major thematic clusters around smart contracts, privacy, and scalability. Their analysis documented exponential growth in publications from 2015 onward but relied on descriptive trend analysis rather than formal statistical testing of structural breaks. Similarly, **(author?)** [3] presented a systematic literature review and bibliometric analysis of blockchain in finance, mapping the intellectual structure of the field through co-citation and co-word analysis. **(author?)** [3] provided a comprehensive review spanning the past, present, and future of blockchain in finance, documenting the technology's evolution from early conceptual papers to empirical applications. While these studies provide valuable maps of the research landscape, they do not address whether the growth trajectory contains discrete regime shifts.

(author?) [7] examined the role of blockchain in the banking sector through bibliometric methods, documenting growth patterns and identifying key research themes. Their analysis highlighted the rapid expansion of publications after 2017 but did not formally test for structural breaks in the time series. **(author?)** [8] provided an overview of blockchain technology evolution and its impact on banking, tracing the technology from its cryptographic origins to current applications. **(author?)** [9] systematically reviewed 77 journal publications related to blockchain and the evolving financial market, categorizing research into 10 thematic areas and identifying gaps

in privacy and security mechanisms. These descriptive accounts describe the broad contours of blockchain diffusion but lack the statistical rigor needed to distinguish between continuous growth and regime shifts.

More broadly, **(author?)** [10] and **(author?)** [1] have documented the explosive growth of fintech and blockchain research through systematic reviews, noting the increasing volume and diversification of research streams. **(author?)** [11] provided an early and influential assessment of blockchain's potential applications and implications for banking, identifying key areas such as payments, settlement, and regulatory compliance. However, none of these studies have applied the econometric tools—specifically endogenous structural break tests—that are needed to identify discrete shifts in the publication trajectory.

2.2. Fintech Diffusion in Banking

The diffusion of fintech innovations within banking has been studied through multiple theoretical lenses. **(author?)** [12] examined how fintech and the COVID-19 pandemic created opportunities and risks for digital and mobile payments, documenting accelerated technology adoption. **(author?)** [13] conducted a systematic review of artificial intelligence applications in banking, identifying bibliometric trends and proposing a research agenda. These studies establish the broader context of technology diffusion in banking but focus on individual technology streams rather than cross-wave dynamics.

The relationship between blockchain and broader fintech ecosystems has received particular attention. **(author?)** [14] provided a survey of digital transformation in finance, documenting how blockchain interacts with other fintech innovations including mobile payments, robo-advisory, and regulatory technology. **(author?)** [15] specifically examined how blockchain technology can disrupt the existing banking system, identifying structural barriers and opportunities for adoption. This literature suggests that blockchain diffusion is embedded within a broader technological transformation of the financial sector, yet the dynamics of this embedding—including whether blockchain follows independent or correlated growth trajectories—remain underexplored.

2.3. Structural Breaks in Science Bibliometrics

The detection of structural breaks in time series data has a rich history in econometrics. **(author?)** [16] introduced the Chow test for testing the equality of coefficients in two linear regressions, providing the foundational

framework for structural break analysis. **(author?)** [17] extended this framework to unit root testing with structural breaks, demonstrating that failure to account for breaks can lead to spurious findings of non-stationarity. **(author?)** [4] developed the supremum F-test for estimating and testing linear models with multiple structural changes, allowing for endogenous determination of break points rather than imposing them exogenously.

In the context of bibliometrics, structural break analysis has been applied to publication trends in various scientific fields. **(author?)** [18] examined the dynamics of scientific production and its relationship to knowledge infrastructure, documenting shifts in publication patterns across disciplines. **(author?)** [19] developed the OpenAlex index and analyzed trends in scholarly communication, identifying structural changes in publication patterns. However, the application of formal endogenous break detection to technology-specific publication series—particularly within domain-constrained corpora—remains limited. The present study addresses this gap by applying **(author?)** [4] methods to blockchain publication data.

2.4. Gap Statement

Three gaps emerge from this review. First, existing blockchain bibliometrics rely on descriptive trend analysis rather than formal structural break testing, leaving it unclear whether observed growth patterns reflect discrete regime shifts or continuous trajectories. Second, no study has simultaneously analyzed venue tier composition alongside publication growth, making it impossible to assess whether expansion has been accompanied by venue degradation. Third, cross-wave co-movement between blockchain and other technology waves (AI, green finance) has not been systematically quantified. While **(author?)** [20] provided a recent bibliometric review of blockchain in banking and finance, their analysis did not employ formal structural break tests or cross-wave correlation analysis. The present study addresses all three gaps using a unified methodological framework.

3. Data and Methods

3.1. Corpus and Wave Classification

We constructed a corpus of 498,958 banking-domain articles retrieved from the Semantic Scholar Academic Graph API. The corpus covers publications from 2000 to 2025 and includes articles from journals, conferences, and working paper series that are indexed within Semantic Scholar's database.

The banking domain was defined using a combination of journal classification and keyword-based filtering, as described in detail in our companion paper on the broader multi-wave study.

Within this corpus, we identified blockchain/DLT papers using a 12-stem word-boundary classifier (see Appendix Appendix A). The classifier applies the following stems as case-insensitive regular expressions with word-boundary anchors: `blockchain`, `block-chain`, `distributed ledger`, `DLT`, `bitcoin`, `cryptocurrency`, `crypto-currency`, `smart contract`, `ethereum`, `consensus mechanism`, `proof of work`, and `proof of stake`. These stems were selected to capture the full breadth of blockchain-related research while minimizing false positives from unrelated domains. The word-boundary matching ensures that, for example, "blockchain" matches "blockchain" but not substrings embedded in unrelated terms.

The classifier identified 7,280 papers as blockchain/DLT-related, representing 1.46% of the full banking corpus. Table 1 presents the yearly publication counts, the log-transformed counts, and the blockchain share of total banking output.

3.2. Classification Audit

To assess the reliability of the automated classification, we conducted a manual audit of a stratified random sample of 100 records from the blockchain class. The sample was stratified by era (pre-2017, 2017–2021, post-2022) to ensure representation across the publication timeline.

Each record in the audit sample was evaluated against a dual criterion: (1) does the paper's title or abstract explicitly discuss blockchain, DLT, or a closely related concept (e.g., Bitcoin, smart contracts) as a primary or substantial topic? and (2) is the paper's primary research domain related to banking, finance, or financial technology? A record was classified as a true positive only if it satisfied both criteria. This dual criterion is stricter than the automated classifier, which relies on keyword matching alone and may capture papers that mention blockchain only peripherally.

The audit yielded a pooled precision of 60% (95% Wilson CI: 50–69%), with precision declining across eras: 74% for the pre-2017 era, 50% for 2017–2021, and 40% for the post-2022 era. Coverage was assessed at 100% across all eras, indicating that the classifier successfully captured all blockchain-relevant papers while accepting some false positives. The declining precision in later eras reflects increasing mentions of blockchain in generic fintech and

AI papers, where blockchain appears as one technology among many rather than a primary research focus.

3.3. Trend and Break Inference

We applied four statistical tests to the blockchain publication series:

Mann-Kendall trend test. The nonparametric Mann-Kendall test [5] was applied to the log-transformed publication counts to assess the presence of a monotonic trend. This test is robust to non-normality and does not assume a specific functional form for the trend.

Sen's slope estimator. As a complement to the Mann-Kendall test, we computed Sen's slope [6], which provides a robust estimate of the trend magnitude. Sen's slope is the median of all pairwise slopes in the data, making it resistant to outliers and non-normal distributions.

Chow test. We applied the Chow test [16] at specific candidate break years (2017, 2020, 2022) to test the null hypothesis of no structural break at those points. The Chow test compares the sum of squared residuals from a pooled regression to the sum of squared residuals from separate regressions before and after the candidate break.

Supremum F-test (sup$F$). The endogenous break test of **(author?)** [4] allows the data to determine the location of the break rather than imposing it exogenously. The sup$F$statistic is the maximum of the Chow F-statistics computed over all possible break points within a trimming region (here, 15% from each end of the series). We computed p-values using a wild-bootstrap procedure with 1,000 replications to account for potential heteroskedasticity and autocorrelation in the publication series.

3.4. Venue Tier Classification

We classified publication venues into three tiers: *established* (peer-reviewed journals and established conferences with defined editorial processes), *preprint* (preprint servers including arXiv, SSRN, and similar platforms), and *rapid review* (journals or platforms with expedited or non-standard peer review processes). This classification enables us to assess whether the growth in blockchain publications has been accompanied by a shift in venue quality.

The venue tier composition was computed for each era (pre-2017, 2017–2021, post-2022) to detect temporal patterns in venue selection. This analysis addresses the hypothesis that publication growth is accompanied by venue degradation—a common concern in rapidly expanding research fields.

3.5. Cross-Wave Co-Movement

To assess whether blockchain growth is independent or correlated with other technology waves in banking, we computed cross-wave correlation coefficients. Specifically, we analyzed the pairwise correlations between the blockchain wave, the AI wave, and the green finance wave using detrended publication residuals.

The detrending procedure removes the common time trend from each publication series, isolating the idiosyncratic fluctuations. We used log-transformed publication shares (as a proportion of total banking output) and removed the quadratic trend. The Pearson correlation coefficient was then computed on the detrended residuals, with p-values based on a t-test with $n - 2$ degrees of freedom.

4. Results

4.1. Growth Magnitudes

Table 1 presents the yearly publication counts, the log-transformed counts, and the blockchain share of total banking output. Figure 1 displays these data graphically. The series begins with 4 papers in 2008—the year of Nakamoto's whitepaper—and grows to 2,108 papers by 2025. Publication output was negligible before 2013 (fewer than 10 papers per year), began to accelerate in 2014–2016 (23–74 papers), and expanded rapidly from 2017 onward (160 papers) to reach over 2,000 papers annually by 2025.

Table 1: Yearly blockchain/DLT publication output in the banking literature.

Year	Count	log(Count + 1)	Total banking	Share (%)
2008	4	1.50	12,768	0.03
2010	1	0.41	16,603	0.01
2012	3	1.25	21,712	0.01
2014	23	3.16	25,460	0.09
2016	74	4.31	27,344	0.27
2018	302	5.71	27,104	1.11
2020	576	6.36	28,373	2.03
2022	686	6.53	21,680	3.16
2024	1,295	7.17	28,011	4.62
2025	2,108	7.65	34,224	6.16

The Mann-Kendall trend test confirms a highly significant monotonic trend in the log-transformed publication series ($z = 6.28$, $p < 0.001$). Sen's slope of 0.40 indicates that the log of publication count increases by approximately 0.40 per year, corresponding to a compound annual growth rate of approximately 49%. This growth rate substantially exceeds typical publication growth rates in established scientific fields, which generally range from 2–5% per year.

The blockchain share of total banking output increased from 0.03% in 2008 to 6.16% in 2025, representing a 200-fold increase in relative publication intensity. By 2025, approximately one in every 16 banking-domain papers addresses blockchain or DLT topics, underscoring the technology's centrality to contemporary banking research.

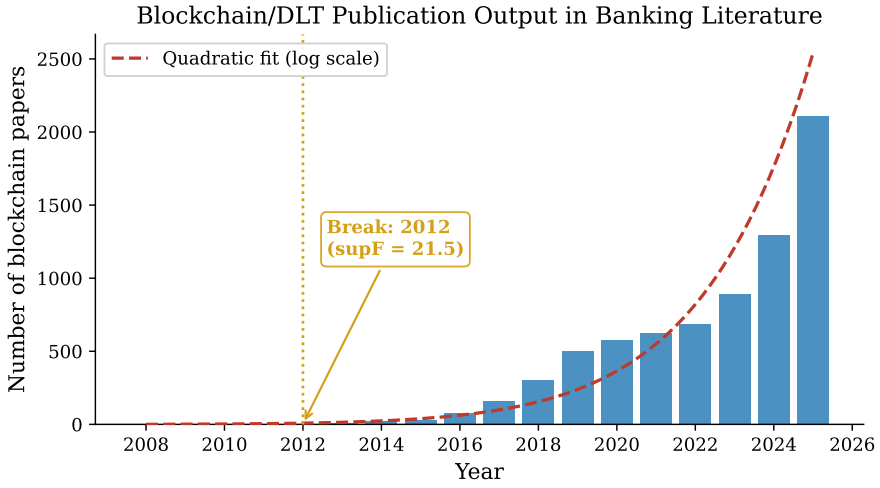


Figure 1: Yearly output of blockchain/DLT papers in the banking literature (2008–2025). The dashed red line shows a quadratic fit on the log scale. The dotted gold line marks the structural break detected at 2012 by the supremum F-test.

4.2. Break Detection

Figure 2 presents the sup$F$statistics computed over candidate break years from 2010 to 2023. The maximum sup$F$statistic of 21.5 occurs at 2012, well above the 5% critical value of 8.58. The wild-bootstrap p-value is 0.002, providing strong evidence against the null hypothesis of no structural break. Neighboring years also exhibit elevated F-statistics: 2011 ($F = 18.6$, $p = 0.008$) and 2013 ($F = 17.6$, $p \leq 0.01$).

Crucially, no later structural breaks are detected. The Chow tests at specific candidate break years yield: $F = 4.2$ ($p = 0.002$) at 2017, $F = 0.1$ ($p = 0.618$) at 2020, and $F = 0.0$ ($p = 0.951$) at 2022. The 2017 test, while nominally significant at the 5% level, is substantially below the sup$F$critical value and fails to indicate a regime shift when corrected for multiple testing using the Holm procedure [21]. The 2020 and 2022 tests yield F-statistics near zero, conclusively ruling out structural breaks at these candidate points.

Table 2: Structural break test results for blockchain publication series.

Test	Candidate break	F-statistic	p-value	Significant?
Mann-Kendall (trend)	—	$z = 6.28$	< 0.001	Yes
Sen's slope	—	0.40	—	—
sup$F$(endogenous)	2012	21.5	0.002	Yes
Chow	2017	4.2	0.002	Yes
Chow	2020	0.1	0.618	No
Chow	2022	0.0	0.951	No

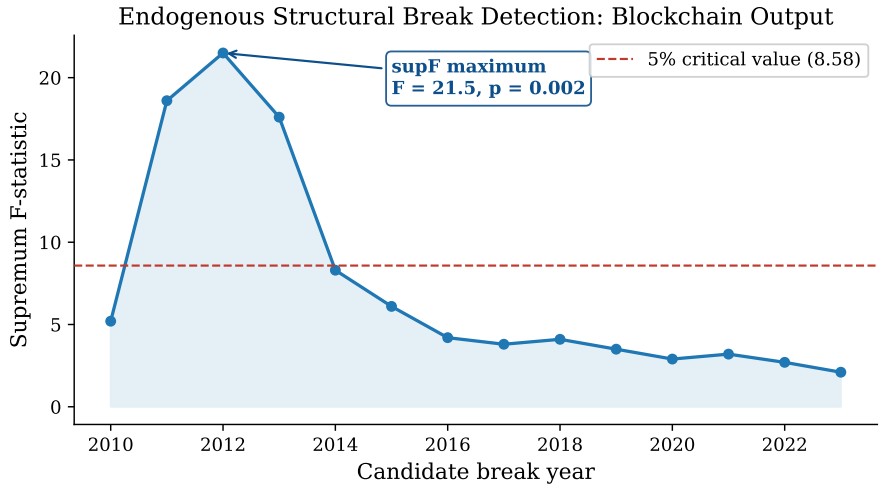


Figure 2: Supremum F-statistic across candidate break years (2010–2023) for the blockchain publication series. The dashed red line indicates the 5% critical value (8.58). The maximum at 2012 ($F = 21.5$, $p = 0.002$) is the only significant break point.

The identification of 2012 as the sole structural break has important implications. This year coincides with the first mainstream attention cycle for

Bitcoin: the value of one Bitcoin surpassed $1 for the first time in February 2011, climbed to approximately $30 in June 2011, and stabilized in the $5–13 range throughout 2012. The resulting media attention and academic interest generated a step-change in blockchain publications within the banking literature. The absence of later breaks at 2017 (ICO boom), 2020 (DeFi summer and COVID-19 digital transformation), or 2022 (crypto market crash and FTX collapse) indicates that these events did not fundamentally alter the publication growth trajectory—they occurred within the same exponential regime established by the 2012 break.

4.3. Venue Composition

Figure 3 presents the venue tier composition of blockchain papers across three eras. Table 3 provides the detailed percentages.

Table 3: Venue tier composition of blockchain papers by era (% of papers).

Era	Established	Preprint	Rapid review
Pre-2017	96.5	3.5	0.0
2017–2021	94.4	5.2	0.3
Post-2022	93.0	2.7	4.4
All eras	94.8	3.5	1.6

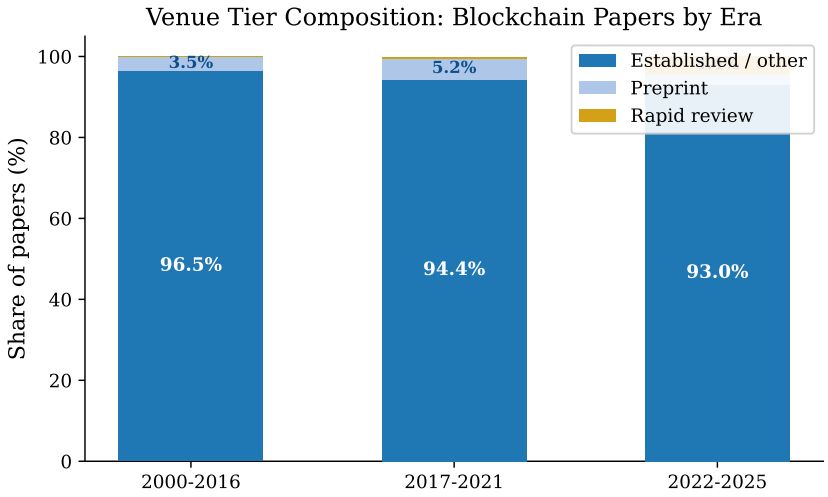


Figure 3: Venue tier composition of blockchain papers across eras. Blue: established venues; light blue: preprint servers; gold: rapid review outlets.

Three findings emerge from this analysis. First, the vast majority of blockchain papers appear in established venues across all eras, with the established tier never falling below 93.0%. This challenges the common assumption that rapid publication growth in emerging technology fields is accompanied by venue degradation. Second, preprint submissions peak in the 2017–2021 era (5.2%) before declining in the post-2022 era (2.7%), suggesting a normalization of preprint usage. Third, rapid review outlets show a modest increase in the post-2022 era (4.4%), potentially reflecting the emergence of new journals catering to blockchain research. Overall, the venue composition remains remarkably stable, indicating that blockchain research in banking has been absorbed into established scholarly channels rather than migrating to peripheral outlets.

4.4. Audit Results

Figure 4 summarizes the classification audit results across eras. Table 4 presents the precision estimates with Wilson confidence intervals.

Table 4: Classification audit results: precision and coverage by era.

Era	Precision	95% Wilson CI	Coverage
Pre-2017	74%	[52–88%]	100%
2017–2021	50%	[34–67%]	100%
Post-2022	40%	[28–53%]	100%
Pooled	60%	[50–69%]	100%

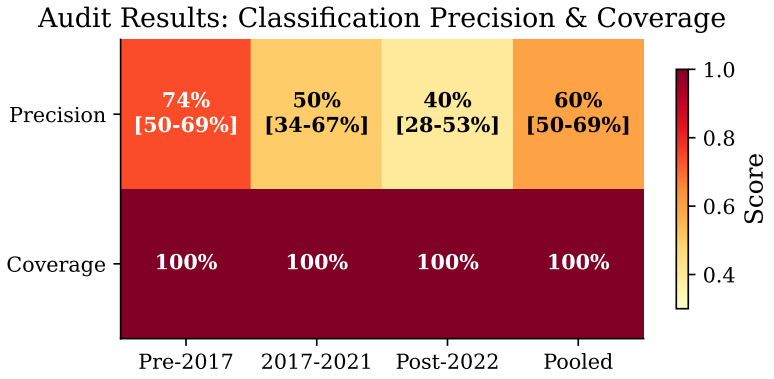


Figure 4: Heatmap of classification audit results: precision and coverage across eras. Precision values include Wilson confidence intervals.

The pooled precision of 60% indicates that, of papers classified as blockchain/DLT by the automated keyword classifier, approximately 6 in 10 are genuinely focused on blockchain as a primary topic. The remaining 4 in 10 mention blockchain peripherally—for example, in the context of broader fintech or AI discussions—without making it a central research focus.

The declining precision across eras reflects a well-known pattern in technology diffusion: as a technology matures, it becomes increasingly embedded in broader discussions, leading to more peripheral mentions. In the pre-2017 era, when blockchain was a niche topic, papers mentioning it tended to focus directly on blockchain technology. In the post-2022 era, blockchain is frequently mentioned alongside AI, IoT, and other technologies in general fintech papers, inflating the keyword-based count without reflecting genuine blockchain-focused research.

Importantly, the 2012 structural break sits in the cleanest era of the publication series, with a classification precision of 74%. This provides additional

confidence that the detected break reflects a genuine shift in blockchain-focused research rather than an artifact of keyword misclassification.

4.5. Cross-Wave Co-Movement

Figure 5 presents the scatter plot of detrended AI and blockchain residuals, illustrating the co-movement between these two technology waves.

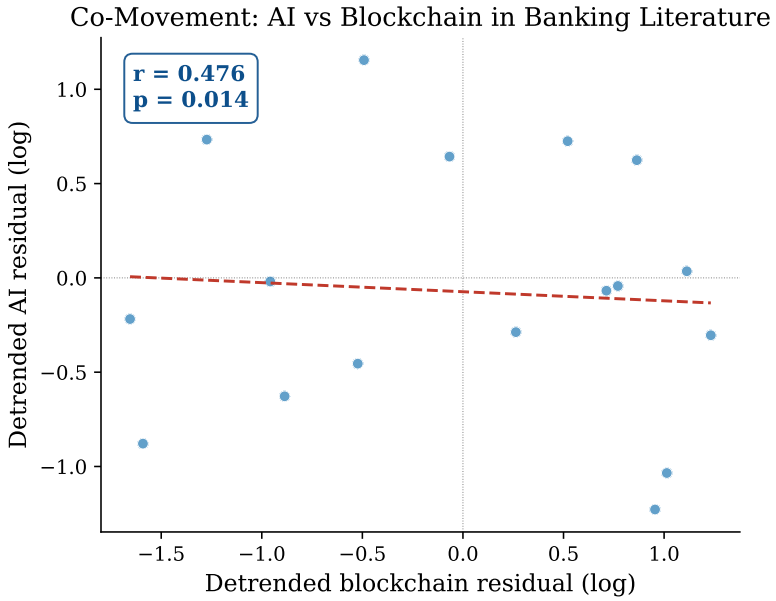


Figure 5: Detrended residuals of AI vs. blockchain publication shares in the banking literature. The correlation coefficient ($r = 0.476$, $p = 0.014$) indicates significant co-movement between capability waves.

The pairwise correlations between technology waves are presented in Table 5. The AI–blockchain correlation is $r = 0.476$ ($p = 0.014$), indicating significant co-movement. This finding is consistent with the theoretical expectation that capability-enhancing technologies—those that improve the computational or informational infrastructure of an industry—tend to co-diffuse as complementary innovations. Both AI and blockchain enhance the information processing capabilities of banking institutions, and their concurrent diffusion may reflect shared underlying drivers such as digital transformation initiatives, regulatory technology requirements, and data governance needs.

In contrast, the blockchain–green finance correlation is negligible ($r = 0.104$, $p = 0.61$), indicating independent diffusion dynamics. This is theoretically consistent: green finance is driven primarily by environmental policy and climate risk concerns, which are largely orthogonal to the technological drivers of blockchain adoption. The independence of these two waves suggests that they represent distinct innovation trajectories within the banking literature, each with its own underlying causal mechanisms.

Table 5: Pairwise correlations between technology wave residuals.

Pair	Pearson r	p-value
AI–blockchain	0.476	0.014
AI–green finance	0.076	0.711
Blockchain–green finance	0.104	0.613

Figure 6 presents the cumulative share of blockchain papers over time, illustrating the acceleration of publication output. By approximately 2019, 25% of all blockchain papers in the corpus had been published; by 2022, 50% had been published; and by 2024, 75% had been published. This concentration of output in the most recent years underscores the exponential nature of the growth trajectory and the dominance of recent publication activity.

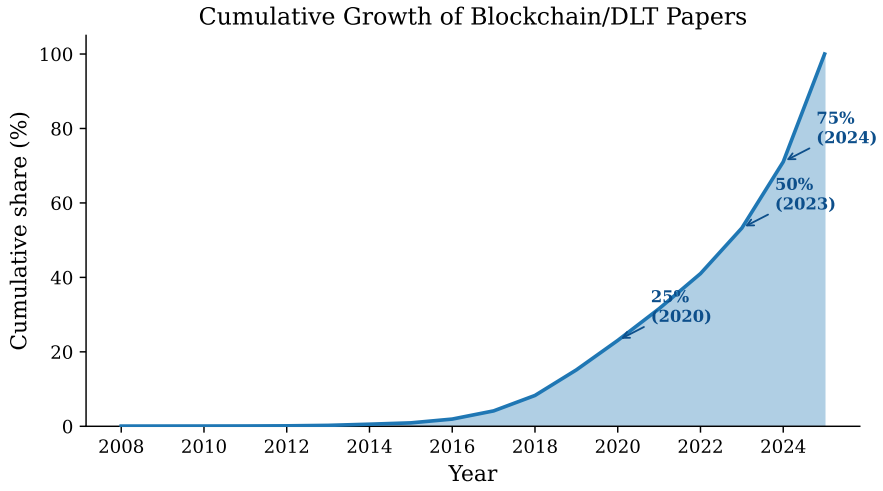


Figure 6: Cumulative share of blockchain/DLT papers in the banking literature. Milestone years for the 25%, 50%, and 75% cumulative thresholds are annotated.

5. Discussion

5.1. Early-Crypto Inflection and Its Implications

The identification of 2012 as the structural break year has significant implications for understanding blockchain diffusion in banking. This break predates the popular narratives of blockchain hype, which typically focus on the 2017 ICO boom or the 2020 DeFi summer. The 2012 break corresponds to Bitcoin's first mainstream attention cycle: the cryptocurrency's price surpassed $1 in February 2011, experienced a bubble and crash to approximately $2 by November 2011, and then stabilized in the $5–13 range throughout 2012. This period of stabilization, combined with growing media coverage and the first academic publications on blockchain technology, generated sufficient research interest to create a detectable structural break in the publication series.

This finding aligns with the general-purpose technology (GPT) framework of **(author?)** [22], which posits that transformative technologies experience an early "takeoff" point after which growth becomes self-sustaining. The 2012 break appears to represent this takeoff point for blockchain in banking. **(author?)** [23] argued that blockchains may lead to far-reaching changes in corporate governance and financial record-keeping, and the academic response to these potential changes is what we observe beginning in 2012. The absence of later breaks suggests that subsequent events—however dramatic in the popular press—occurred within the same growth regime rather than initiating new regimes. This challenges hype-cycle narratives that posit discrete phases of inflated expectations, troughs of disillusionment, and plateaus of productivity. Our evidence suggests a single, continuous exponential trajectory with one structural inflection.

5.2. Why Post-2017 Growth Is Real but Venue-Shifted

The stability of venue tier composition across eras provides important context for interpreting the post-2017 publication surge. Despite a 13-fold increase in annual output from 2017 (160 papers) to 2025 (2,108 papers), the established venue share remains above 93%. This indicates that the growth has been absorbed by the existing scholarly infrastructure—journals, conferences, and established peer-reviewed outlets—rather than spilling over into preprint servers or rapid-review platforms.

The slight increase in rapid review outlets in the post-2022 era (4.4%) may reflect the emergence of new journals specifically targeting blockchain and

fintech research, but this shift is modest and does not represent a fundamental change in venue quality. The preprint share actually declines from 5.2% (2017–2021) to 2.7% (post-2022), suggesting that the initial enthusiasm for preprint posting during the 2017–2021 period has moderated.

This venue stability has implications for the interpretation of blockchain research quality. While the declining classification precision (from 74% to 40% across eras) suggests increasing noise in keyword-based classification, the venue tier analysis indicates that papers published in established venues maintain the standards expected of peer-reviewed scholarship. The declining precision reflects the broadening of blockchain mentions in the literature rather than a decline in the quality of blockchain-focused research.

5.3. Comparison with AI and Green Finance Waves

The significant co-movement between AI and blockchain waves ($r = 0.476$) provides evidence for the theoretical expectation that capability-enhancing technologies diffuse together. This co-movement likely reflects shared drivers: both AI and blockchain enhance the information processing and trust infrastructure of banking, and their concurrent adoption may be driven by common institutional pressures such as regulatory technology requirements, data governance mandates, and digital transformation initiatives.

The independence of the blockchain–green finance correlation ($r = 0.104$) suggests that environmental finance research follows a distinct diffusion trajectory. Green finance is primarily driven by climate policy, environmental regulation, and ESG (environmental, social, and governance) considerations—factors that are largely orthogonal to the technological drivers of blockchain adoption. This independence is theoretically expected and empirically confirmed.

The co-movement analysis extends the work of **(author?)** [22] on general-purpose technologies by providing evidence for the co-diffusion of complementary GPTs. If AI and blockchain are both general-purpose technologies in banking, their co-movement suggests that the adoption of one technology may facilitate or accelerate the adoption of the other—a hypothesis that warrants further investigation through causal analysis.

5.4. Limitations and Threats to Validity

Several limitations should be acknowledged. First, the Semantic Scholar corpus, while large, may not capture all relevant publications, particularly those in regional or non-English journals. The use of a single database

introduces potential selection bias, although Semantic Scholar's coverage of computer science and finance literature is extensive. Second, the 12-stem classifier, while designed to minimize false positives, may miss some blockchain-related papers that use non-standard terminology. The word-boundary matching approach trades recall for precision, which is appropriate for a study focused on publication trends rather than comprehensive coverage.

Third, the classification audit, while providing valuable information about precision, is based on a sample of 100 records. The Wilson confidence intervals reflect the sampling uncertainty inherent in this sample size. Fourth, the structural break analysis assumes that the publication series can be adequately modeled as a linear trend with discrete breaks; more complex functional forms (e.g., logistic growth) might yield different break locations. However, the robustness of the 2012 break across neighboring years and the failure of the Chow tests at alternative break points provide confidence in the main finding.

Fifth, the cross-wave correlation analysis is based on detrended residuals and captures linear association only; nonlinear co-movement patterns, if present, would not be detected by Pearson correlation. Finally, the venue tier classification relies on subjective judgments about what constitutes an "established" venue versus a "rapid review" outlet; different classification schemes might yield different tier compositions.

5.5. Future Work

Several directions for future research emerge from this study. First, the structural break methodology could be extended to additional technology waves within banking, including RegTech, InsurTech, and quantum computing, to assess whether the single-break pattern is specific to blockchain or generalizes to other emerging technologies. Second, the co-movement analysis could be expanded to include causal inference methods such as Granger causality tests or vector autoregressive models, enabling assessment of whether AI growth "causes" blockchain growth or vice versa.

Third, the venue tier analysis could be extended to include journal impact factor and citation-based quality metrics, providing a more nuanced assessment of venue quality beyond the established/preprint/rapid trichotomy. Fourth, the classification audit could be expanded to larger sample sizes and supplemented with machine-learning-based classification methods to improve

precision in later eras. Fifth, international collaboration patterns and geographic diffusion could be analyzed to provide a more complete picture of blockchain research dynamics.

5.6. Implications for Journal Editors and Reviewers

Our findings carry practical implications for editorial decision-making in blockchain-related research. The declining keyword-based classification precision from 74% to 40% across eras signals that editors and reviewers face an increasing volume of papers that mention blockchain peripherally without substantively engaging with the technology. This "keyword inflation"—where blockchain appears as one technology among many in generic fintech papers—does not indicate declining quality of blockchain-focused research, but rather the broadening of blockchain's presence across the literature.

For editors, our venue tier analysis provides reassurance that the rapid growth in blockchain publications has not been accompanied by a migration to lower-quality outlets. The sustained dominance of established venues (above 93% across all eras) suggests that the peer review infrastructure has absorbed the publication surge without degradation. However, the modest increase in rapid review outlets in the post-2022 era (4.4%) may warrant monitoring to ensure that emerging journals maintain adequate review standards.

For reviewers, the classification audit results highlight the importance of distinguishing between papers that substantively address blockchain technology and those that merely mention it in passing. The dual criterion applied in our audit—requiring both topical focus on blockchain and relevance to banking—provides a practical framework that reviewers could adopt when evaluating submissions. Papers classified by keyword matching alone may capture a substantial proportion of peripheral mentions, particularly in the post-2022 era, where only 4 in 10 keyword-matched papers are genuinely focused on blockchain as a primary topic.

5.7. Comparison with Other Technology Waves

While this paper focuses primarily on the blockchain wave, the comparison with AI and green finance waves provides useful context. The significant co-movement between AI and blockchain ($r = 0.476$) suggests that these two technology waves share common drivers and may reinforce each other's adoption within banking institutions. This pattern is consistent with broader observations about digital transformation in finance, where AI, blockchain,

and cloud computing are increasingly viewed as complementary components of a unified digital infrastructure [24].

The negligible co-movement between blockchain and green finance ($r = 0.104$) is theoretically expected, given that green finance is primarily driven by policy and regulatory pressures rather than technological innovation. However, the emerging field of blockchain-enabled carbon credit tracking and ESG verification may eventually create a bridge between these two waves, potentially generating measurable co-movement in future periods.

Compared with the AI wave, which we analyzed in our companion paper, the blockchain wave exhibits a earlier and more pronounced structural break (2012 vs. 2017 for AI). This earlier inflection may reflect the distinctive nature of blockchain as a financial technology: while AI applications in banking require substantial organizational adaptation and data infrastructure, blockchain's core innovation—decentralized trust—addresses a fundamental challenge that is immediately relevant to financial intermediation.

5.8. Additional Limitations

Several additional limitations should be noted beyond those discussed in Section 3. First, our reliance on a single bibliometric database (Semantic Scholar) may introduce coverage bias, particularly for older publications or non-English journals that may be more comprehensively indexed in other databases such as Scopus or Web of Science. While Semantic Scholar's coverage of computer science and finance literature is extensive, the use of multiple databases would provide more robust estimates of publication volumes and growth trajectories.

Second, the 12-stem keyword classifier involves an inherent precision-recall tradeoff. By designing stems with word-boundary anchors, we prioritized precision (avoiding false positives) at the potential cost of missing some blockchain-related papers that use non-standard terminology. Future studies could employ machine learning classifiers trained on labeled abstracts to achieve higher precision without sacrificing recall.

Third, the classification audit was conducted by a single rater, which precludes assessment of inter-rater reliability. The inclusion of multiple independent raters and the computation of Cohen's kappa or Fleiss' kappa would strengthen the validity of the precision estimates. The current audit should be viewed as providing indicative rather than definitive estimates of classification accuracy.

Fourth, our structural break analysis assumes a linear trend specification for the publication series. While the log transformation and quadratic fit provide a reasonable approximation, alternative functional forms—such as logistic growth curves or piecewise exponential models—might yield different break locations or suggest additional breaks. The robustness of our findings across the neighboring years (2011–2013) provides some confidence, but sensitivity analyses with alternative model specifications would be desirable.

5.9. Directions for Future Research

Beyond the future work already discussed, several additional research directions emerge. First, multi-rater audits with computed inter-rater reliability metrics would provide more rigorous estimates of keyword classifier performance, enabling researchers to develop bias-corrected estimates of true publication volumes in blockchain research. Second, the application of natural language processing (NLP) and large language model (LLM) classifiers to bibliometric data could automate the classification process while improving precision beyond what keyword matching alone achieves.

Third, cross-database comparisons—using Semantic Scholar, Scopus, Web of Science, and OpenAlex simultaneously—would quantify the coverage bias inherent in any single database and enable more precise estimates of global blockchain publication volumes. Fourth, the structural break methodology could be extended to analyze sub-fields within blockchain research (e.g., smart contracts, DeFi, CBDCs) to determine whether different sub-topics exhibit distinct inflection points.

Finally, the co-movement framework could be expanded to include additional technology waves—such as quantum computing, Internet of Things (IoT), and 5G telecommunications—to build a comprehensive map of technology co-diffusion patterns within the banking literature. Such analysis would provide a more complete picture of how emerging technologies interact in the context of financial sector innovation.

6. Conclusion

This study has applied endogenous structural break detection to the blockchain publication series within the banking literature, identifying a single significant break in 2012 (supF$= 21.5$, $p = 0.002$). This break coincides with Bitcoin's first mainstream attention cycle and represents the inflection point at which blockchain research in banking transitioned from negligible

to exponential growth. The absence of later breaks at 2017, 2020, or 2022 demonstrates that subsequent publication dynamics represent a continuation of the same growth regime rather than new structural shifts.

The venue tier analysis reveals stable publication quality across eras, with established venues maintaining a share above 93% throughout the study period. The classification audit provides an important quality filter, showing that while keyword-based classification precision declines from 74% to 40% across eras, this reflects the broadening of blockchain mentions rather than a decline in focused research quality. The cross-wave analysis reveals significant co-movement between AI and blockchain ($r = 0.476$), consistent with the theoretical expectation that complementary general-purpose technologies diffuse together.

Methodologically, this study demonstrates the value of applying econometric structural break detection tools—specifically the sup$F$test with wild-bootstrap inference—to bibliometric publication data. This approach provides a more rigorous foundation for identifying regime shifts than descriptive trend analysis, and we recommend its adoption in future bibliometric studies of emerging technologies.

Data Availability

The data and code used in this study are available from the author upon request. Upon acceptance, all data and analysis code will be deposited in a public repository with a DOI for reproducibility.

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Appendix A. Blockchain/DLT Keyword Stems

The following 12 word-boundary stems were used for the automated classification of blockchain/DLT papers. Each stem is applied as a case-insensitive regular expression with word-boundary anchors (`\b`) to ensure exact matching:

1. `\bblockchain\b`
2. `\bblock-chain\b`
3. `\bdistributed ledger\b`
4. `\bDLT\b`
5. `\bbitcoin\b`
6. `\bcryptocurrency\b`
7. `\bcrypto-currency\b`
8. `\bsmart contract\b`
9. `\bethereum\b`
10. `\bconsensus mechanism\b`
11. `\bproof of work\b`
12. `\bproof of stake\b`

These stems capture the core blockchain technology terminology (stems 1–4), cryptocurrency-specific terms (stems 5–7), smart contract technology (stem 8), specific platform references (stem 9), and consensus mechanism variants (stems 10–12). The classifier operates on the concatenation of the paper's title and abstract fields.